List of publications by Jorge Alda

Note: All citation figures are given by Inspire-HEP as of October 12, 2026, and exclude self-citations.

1 Published articles

J. Alda, J. Guasch and S. Peñaranda: ***Some results on Lepton Flavour Universality Violation***
Eur.Phys. J. C, 79 7 (2019) 588
doi:10.1140/epjc/s10052-019-7092-x
arXiv:1805.03636 [hep-ph]
15 citations.

J. Alda, J. Guasch and S. Peñaranda: ***Anomalies in B decays: A phenomenological approach***
Eur.Phys. J. Plus 137 (2022) 217
doi:10.1140/epjp/s13360-022-02405-3
arXiv:2012.14799 [hep-ph]
14 citations.

J. Alda, J. Guasch and S. Peñaranda: ***Using Machine Learning techniques in phenomenological studies in flavour physics***
J. High Energ. Phys. 2022, 115 (2022)
doi:10.1007/JHEP07(2022)115
arXiv:2109.07405 [he-ph]
4 citations.

J. Alda, A. W. M Guerrera, S. Peñaranda and S. Rigolin: ***Leptonic Meson Decays into Invisible ALP***
Nucl.Phys.B 979 (2022) 115791
doi:10.1016/j.nuclphysb.2022.115791
arXiv:2111.02536 [hep-ph]
24 citations.

J. Alda, G. Levati, P. Paradisi, S. Rigolin, N. Selimović: ***Collider and astrophysical signatures of light scalars with enhanced τ couplings***
JHEP 06 (2025) 008
doi:10.1007/JHEP06(2025)008
arXiv:2407.18296 [hep-ph]
27 citations.

S. Peñaranda, J. Alda, A. Mir: ***Flavour Anomalies: A comparative Analysis using a Machine Learning Algorithm***
Int.J.Theor.Phys. 65 (2026) 2, 46
doi:10.1007/s10773-026-06263-y

arXiv:2412.15830 [hep-ph]
2 citations.

J. Alda, C. Broggini, G. Di Carlo, L. Di Luzio, D. Piatti, S. Rigolin, C. Toni: ***Weak nuclear decays deep-underground as a probe of axion dark matter***
Phys.Rev.D 111 (2025) 3, 035022
doi:10.1103/PhysRevD.111.035022
arXiv:2412.20932 [hep-ph]
16 citations.

J. Alda, J. Asorey, A. Mir, S. Peñaranda: ***Physically Consistent Parameter Inference: Transparent Machine Learning Emulation in High Energy Physics and Cosmology***
Universe. 2026; 12(8):238
doi:10.3390/universe12080238
arXiv:2607.12726 [hep-ph]
Invited contribution to special issue "Advances in Machine Learning Techniques in Cosmology and Particle Physics".
1 citation.

2 Pre-prints

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz, S. Rigolin: ***Comprehensive ALP Searches in Meson Decays***
arXiv:2507.19578 [hep-ph]
Submitted to Fortschritte der Physik.
18 citations.

J. Alda, M. Fuentes Zamoro, L. Merlo, X. Ponce Díaz, S. Rigolin: ***ALPaca: The ALP Automatic Computing Algorithm***
arXiv:2508.08354 [hep-ph]
Submitted to Computer Physics Communications.
7 citations.

J. Alda, S. Rigolin: ***ALP interactions with mesons and baryons at $\mathcal{O}(1/f_a)$: a general field redefinition invariant approach***
arXiv:2610.13671 [hep-ph]

3 Proceedings

J. Alda, J. Guasch and S. Peñaranda: ***Anomalies in B decays: Present status and future collider prospects***
arXiv:2105.05095 [hep-ph]
SLAC eConf C21-03-15.1.
Proceedings to LCWS 2021.
4 citations.

J. Alda, J. Guasch and S. Peñaranda: ***Exploring B-physics anomalies at colliders***
arXiv:2110.12240 [hep-ph]
doi:10.22323/1.398.0494
Proceedings to EPS-HEP2021.
1 citation.

A. Mir, J. Alda, S. Peñaranda: ***B-Meson Anomalies: Effective Field Theory Meets Machine Learning***
doi:10.22323/1.485.0349
arXiv:2510.17742 [hep-ph]
Proceedings to EPS-HEP2025.

M. Fuentes Zamoro, J. Alda, L. Merlo, X. Ponce Díaz, S. Rigolin: ***Tackling ALP searches in meson decays with ALPaca: a phenomenological approach***
doi:10.22323/1.507.0057
Proceedings to 3rd Training School of the COST Action: Cosmic WISPers (CA21106).

4 Lecture Notes

J. Alda: ***Lecture notes on Machine Learning applications for global fits***
SciPost Phys. Lect. Notes 135 (2026)
doi:10.21468/SciPostPhysLectNotes.135
arXiv:2604.07520 [hep-ph]